

Web Services and Application Big Project

*Worth 60% of overall marks
(This is marked out of 100)*

Description:

Create a Web application that uses RESTful APIs to perform CRUD operations to some data in one or more database tables, and/or some third party API. I will allow a lot of flexibility in this project, so that you can use it as an opportunity to do something that is useful for your work.

Handup:

A link to the GitHub repository that contains the project (and only this project). The repository should contain:

1. Your code.
2. A "ReadMe" file if there is anything I need to do to run this code.
3. A hosted version of the web application that I can run (Azure, Pythonanywhere) but if you do, please provide the link on your readme.
4. Any other documentation you feel is appropriate.

Deadline:

The deadline is Thursday 14 May 2026 (9:00am), I can not give any extensions to this, because I have to have all the projects corrected and entered into the system by the next week. So pretend that the deadline is earlier (say 1 May and give yourself an extension to the 14 May 😊)

Assessment strategy:

You have flexibility as to what you do for this project. I understand that this can cause confusion as to what you should do, so here schedule of some of the features you may want to implement.

	Description
Minimum project	A rehash of the sample project lab ie A basic hosted Flask server that has a 1. REST API, (to perform CRUD operations) 2. One database table and 3. Accompanying web interface, to perform these CRUD operations. (eg using AJAX calls)
	Features you can add for more marks
	Make it look nice
	Have more than one database table
	Link to an outside data source and either: • Save the data into a database table that you view elsewhere in your application. • Manipulate/analyse the data in real time. I am aware that this can cause issues on some free hosting sites (pythonanywhere).
	Implement user authentication
	Perform more than CRUD operations on your data

Project should be well laid out and easy for me to run.

Marks may be deducted for:

- Poorly formatted code, that I find hard to read (Do not over comment your code), yes you may use AI/LLMs to help you create your code, but the code should be easy to read, understand and maintain
- If I find it hard to run,
- I find it hard to understand your GitHub layout. (a README file is handy).

Alternate project:

I will allow a lot of flexibility with this project, so if you have project in particular you would like to implement, please contact me and I will happily consider it.

This project should:

- Demonstrate you understand APIs.
- Have a nice clear user interface.

The marking scheme is a lot more open on this.

If the data you are dealing with is sensitive and so you cannot give me access to it, you may make a video of you demonstrating your application working (with the sensitive information blurred out)

Research:

Using outside sources is encourage, but you should reference any and all of them.

If you are using an AI system provide the prompts that you used.

Consistency:

There are no specific marks given for consistency, but I will look at your commit history to get a feel about how you went about this project, and I reserve the right to arrange a meeting for you to explain every aspect of your project if I feel it is necessary, or I am just interested in learning more about your project.

Best of Luck.

Email me if you have any questions.